

Waste2Worth: A Human-Computer Interaction Approach to Smart Recycling and Reward-Based Waste Management Platform

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Abstract— Recycling is an important step toward reducing environmental harm, yet many communities still struggle with low participation rates. The main reasons are the lack of motivation, limited access to recycling services, and the absence of a clear and easy process for users to follow. This paper presents Waste2Worth, a mobile application that addresses this problem by combining smart technology with a reward-based system. Users locate nearby smart collection stations, submit recyclable materials, and automatically earn points based on the weight of what they deposit. These points can be redeemed for rewards such as cash or coupons, giving users a direct reason to recycle. The application was developed following a User Centered Design, going through multiple stages from paper prototypes to high-fidelity screens built in Figma. Gamification features such as challenges, badges, and leaderboards were added to keep users engaged over time. Usability evaluation with 23 participants confirmed that the final design is easy to use and well received by users. Future work includes adding AI for waste classification, using blockchain for reward transparency, and testing the system in real-world settings.

Index Terms— Waste Management, Recycling, Mobile Application, Gamification, IoT, User-Centered Design, HCI, Sustainability.

I. INTRODUCTION

Waste management is becoming a critical global issue due to rapid population growth and increasing daily waste production. Traditional collection methods often lack efficiency, relying on fixed schedules and manual labor, which leads to higher operational costs and missed recycling opportunities. A primary challenge remains the lack of user motivation and limited access to convenient recycling services. While recent IoT and mobile solutions have improved logistics, sustaining long-term user engagement continues to be a major hurdle. To address these gaps, this project proposes Waste2Worth, a mobile platform designed to make recycling both accessible and rewarding. The methodology follows a UCD approach, integrating IoT-based smart collection stations with

gamification elements to ensure a seamless and engaging experience. The system allows users including individuals, students, and organizations, to submit materials, earn points, and redeem rewards, thereby fostering sustainable habits. By combining simple technology with a reward-based incentive model, Waste2Worth aims to significantly increase community participation. The remainder of this paper is organized as follows: Section II reviews related literature, Section III details the methodology and system design, Section IV presents the results, followed by the discussion in Section V and concluding remarks in Section VI.

II. LITERATURE REVIEW

A. IoT-Based Smart Waste Management Infrastructure

Several studies have explored the use of IoT technologies as a backbone for modern waste management systems. In [1], a systematic review of over 100 studies revealed that sensor-based bin monitoring, real-time data transmission, and cloud-based analytics significantly reduce collection costs and improve operational efficiency. Expanding on this, [2] proposed a smart city waste solution using IoT sensors and cloud computing to optimize collection routes, achieving measurable reductions in carbon emissions. While both works demonstrate strong infrastructure capabilities, neither incorporated user-facing engagement features or incentive mechanisms to encourage citizen participation. Waste2Worth builds on this IoT foundation by connecting smart collection stations to a mobile interface, while addressing the user engagement gap through a reward and gamification layer.

B. Reward-Based Incentive Systems

A distinct line of research has examined the motivational impact of reward mechanisms on recycling behavior. In [5], a sensor-integrated garbage monitoring system with

embedded reward distribution achieved a 34% increase in correct waste disposal rates among rewarded participants. In [3], tiered unlocking reward strategies were found to sustain pro-environmental behavior over extended periods by combining intrinsic motivation with tangible incentives. In [8], a user-centric digital waste management platform highlighted that transparent reward tracking and achievable milestones are critical for sustaining long-term user engagement. Across these three studies, the common limitation is that rewards functioned in isolation, without integration into a broader gamified ecosystem. Waste2Worth addresses this by combining a points-based reward model with challenges, streaks, and a redeemable voucher system, creating a more complete motivational loop.

C. Gamification and Behavioral Engagement

The application of game mechanics to environmental behavior has been examined from both design and cognitive perspectives. In [6], a gamified recycling station applying scoring, progression levels, and social comparison significantly increased recycling frequency and user satisfaction, particularly among younger users. Complementing this design perspective, [7] examined the neurocognitive mechanisms behind environmental behavior, finding that cue-reward loops and positive reinforcement are among the most effective strategies for building habitual sustainable actions. While [6] focused on physical station-based gamification and [7] remained theoretical, Waste2Worth translates both insights into a mobile context through daily challenges, streak tracking, badge progression, and leaderboard features that operate entirely within the application.

D. Mobile Recycling Applications

Studies examining mobile applications as recycling tools reveal a consistent pattern of partial feature coverage. In [13], WasteApp provided personalized recycling guidance and location-based collection point suggestions for smart city citizens, demonstrating that geolocation features substantially improve correct disposal rates. In [12], an analysis of green mobile applications found that real-time feedback and gamified elements significantly increased recycling rates among consumers. In [17], an effectiveness assessment of smart waste recycling applications confirmed that clear task flows and visible progress tracking are the strongest predictors of user retention. Each of these platforms addressed individual components of the recycling experience, yet none integrated the full spectrum of submission, location, rewards, and challenges within a single unified interface. Waste2Worth consolidates all these functionalities into one cohesive application, eliminating the need for users to switch between multiple tools.

E. Usability Evaluation and Persuasive Design

User-centered evaluation and persuasive design principles have been identified as critical factors in the adoption of sustainability applications. In [9], a usability assessment of a waste bank management application using the System Usability Scale (SUS) identified interface simplicity, clear navigation, and responsive feedback as the primary drivers of

user adoption. In [15], a review of persuasive technology applications across sustainability domains found that personalized goal-setting, progress monitoring, and social influence mechanisms achieve significantly higher behavioral compliance than generic approaches. While [9] focused exclusively on usability measurement without addressing engagement strategies, and [15] remained a theoretical review without a deployed system, Waste2Worth integrates both dimensions by applying HCI-driven design principles throughout development and incorporating persuasive features such as personal impact dashboards and adaptive challenge difficulty.

F. Digital Platforms for Sustainability and Waste Management

A broader set of studies has examined the role of digital platforms in promoting sustainable behavior. In [10], an evaluation of digital tools for eco-conscious behavior promotion found that platforms combining educational content with interactive features were significantly more effective than passive information systems. In [16], a study on digital technologies for waste management across smart city contexts demonstrated that citizen-empowering mobile interfaces can effectively bridge infrastructure gaps. In [14], an integrative review of digitalization in waste recycling identified a growing trend toward unified platforms consolidating multiple recycling functionalities. These studies collectively confirm the market and behavioral need for an integrated application; however, none produced a deployed system combining IoT connectivity, gamification, and rewards. Waste2Worth directly fills this gap.

G. Blockchain and Emerging Transparency Technologies

Blockchain has been proposed as a means of enhancing trust and accountability in waste management. In [18], a comprehensive survey of blockchain-based smart city waste systems demonstrated the technology's potential for improving traceability and reducing fraud in collection and reward processes. In [19], a blockchain-enabled VANET system for solid waste management showed measurable improvements in logistics optimization and operational transparency. Both studies highlight trust and auditability as critical user concerns. While blockchain integration is beyond the current scope of Waste2Worth, these findings informed the design of the visible reward tracking and submission confirmation features, which address user trust through interface transparency rather than distributed ledger technology.

H. Artificial Intelligence in Waste Classification

Two studies examined AI and machine learning as tools for automating waste management. In [4], WasNet, a neural network-based system, achieved high accuracy in classifying waste types and predicting collection schedules on real-world datasets. In [20], a review of AI and machine learning applications in smart waste management identified automated waste classification and predictive analytics as the most impactful areas for improving recycling systems. Both works demonstrate significant potential but require substantial technical infrastructure not available at the

prototype stage. Waste2Worth currently relies on manual waste type selection by users; however, these findings identify automated AI-based classification as a clear and well-supported direction for future development.

I. Barriers to Recycling Adoption

In [11], a state-of-the-art review of plastic solid waste recycling identified three primary barriers to participation: insufficient user awareness, inconvenient collection infrastructure, and lack of motivation. This work provides the most direct justification for Waste2Worth's design rationale, as the application specifically targets all three barriers through its educational impact dashboard, location-based station discovery, and reward-based incentive system respectively.

J. Summary and Research Gap

The reviewed literature collectively demonstrates substantial progress across the individual dimensions of smart waste infrastructure, behavioral incentives, gamification, mobile application design, and usability evaluation. Table I presents a comparative analysis of key features across the most closely related studies. As the table illustrates, no existing work fully integrates IoT connectivity, reward-based incentives, gamification, and location-based

services, and HCI-driven usability evaluation within a single mobile platform. Waste2Worth addresses this gap by unifying all these components into one citizen-centered application, offering a more complete and engaging recycling experience than any previously reviewed system.

Feature	[1]	[2]	[5]	[6]	[13]	[17]	Waste2Worth
IoT Integration	✓	✓	✓	✗	✗	✗	✓
Reward System	✗	✗	✓	✗	✗	✗	✓
Gamification	✗	✗	✗	✓	✗	✗	✓
Location / Map	✗	✓	✗	✗	✓	✗	✓
Mobile App	✗	✗	✗	✗	✓	✓	✓
Usability Testing	✗	✗	✗	✗	✗	✓	✓
HCI / UCD	✗	✗	✗	✗	✗	✗	✓
Unified Platform	✗	✗	✗	✗	✗	✗	✓

TABLE I. Comparison of Related Works.

III. METHODOLOGY AND MATERIALS

A. System Overview

Waste2Worth is a mobile application developed as part of this project to support sustainable waste management. The system encourages users to recycle by offering a reward-based mechanism, where users can submit recyclable materials through smart collection stations and earn points. These points can later be redeemed for rewards such as cash,

coupons, or volunteer hours. This design was derived from the project proposal and refined through earlier development phases.

B. Design Framework

The design of Waste2Worth follows a Human-Computer Interaction (HCI) approach that integrates UCD, gamification, and Internet of Things (IoT) technologies.

The UCD approach was applied during the design process to ensure that the system aligns with user needs and provides a clear and simple interface. The focus was on improving usability, reducing user effort, and enhancing navigation.

Gamification elements such as points, badges, levels, and challenges were incorporated to enhance user engagement. This approach was supported by previous studies showing that gamification improves participation in sustainability-related applications [6].

IoT technology was integrated through smart collection stations that automatically measure the weight of recyclable materials. This concept was based on findings from related studies on smart waste management systems [1], which highlight the effectiveness of automation and real-time data collection.

The system workflow shown in Fig. 1 illustrates the user journey from registration to reward redemption through the Waste2Worth platform.



Fig. 1. Waste2Worth system workflow.

C. System Architecture

The system architecture was defined during the design phase and consists of multiple integrated components.

The mobile application serves as the main interface where users can register, log in, locate nearby collection stations, and track their rewards. Smart collection stations equipped with sensors measure the weight of recyclable materials and transmit this data to the system.

The system processes the data and calculates reward points automatically. This type of integration is supported by IoT based systems that enable real-time monitoring and efficient waste management [1].

Additionally, a reward management module tracks user activities and updates their points. Location-based services are used to identify nearby stations.

D. System Workflow

The system workflow defines the steps users follow when interacting with the application. First, users create an account and log in. After enabling location services, nearby collection stations are displayed on an interactive map. Users then select a station and submit recyclable materials.

The smart station measures the weight and sends the data to the system, where points are calculated and added to the user's account in real time. Users can also participate in environmental challenges to earn additional points. Accumulated points can be redeemed for rewards such as cash, coupons, or volunteer hours. This workflow is consistent with modern automated waste management systems that rely on real-time processing and feedback [2].

E. Implementation Process

The implementation process was carried out in multiple stages as part of the project development phases. Initially, low-fidelity paper prototypes were created to define the layout and interaction flow. These prototypes were evaluated and refined based on user feedback collected during the testing phase. After that, high-fidelity prototypes were developed using Figma to create a more detailed and interactive system design. The overall implementation process is illustrated in Fig. 2.

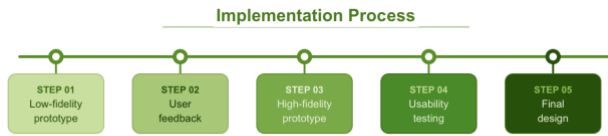


Fig. 2. Waste2Worth implementation process.

Several iterations were performed to improve usability, consistency, and overall user experience, following a UCD approach.

The evaluation process involved 23 participants from different target groups, including students, individual users, and organization representatives, as illustrated in Fig.3. The evaluation combined usability testing and heuristic evaluation to assess usability, navigation, task completion, and overall user experience.

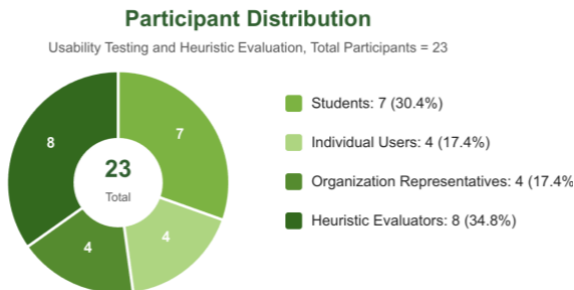


Fig. 3. Participant Distribution for Usability Testing and Heuristic Evaluation.

The usability evaluation also included a System Usability Scale (SUS) assessment to measure overall user satisfaction and interface usability.

The heuristic evaluation was conducted using Nielsen's Ten Usability Heuristics to identify usability issues related to interface consistency, navigation, feedback visibility, and error prevention. Participants evaluated the system interface based on usability principles and provided feedback regarding usability strengths and weaknesses.

In addition, usability testing sessions were conducted using predefined task scenarios, including account creation, locating nearby recycling stations, submitting recyclable materials, completing environmental challenges, and redeeming rewards.

After completing the tasks, participants filled out the System Usability Scale (SUS) questionnaire, which consists of ten standardized usability statements rated on a five-point Likert scale. The final SUS score was calculated based on the standard SUS scoring method to measure overall user satisfaction and interface usability.

F. System Features

The system features were defined based on the project proposal and refined through the design and testing phases. Waste2Worth provides several key functionalities that support both usability and engagement. Users can submit recyclable materials through smart stations, where the system automatically measures weight. An interactive map allows users to locate nearby stations.

The reward system enables users to earn and redeem points, while environmental challenges encourage sustainable behaviors. Additionally, a pickup service is available for organizations handling large quantities of recyclable materials.

G. Materials and Tools

The development of Waste2Worth utilized several tools and technologies during the design and implementation process. Figma Desktop (Version 126.3.12) was used to design both low-fidelity and high-fidelity prototypes, supporting iterative development and usability improvements.

The system also relies on IoT-based smart collection stations equipped with weight sensors, which were identified as an effective solution in smart waste management systems [1].

In addition, location-based services were integrated to help users identify nearby collection stations. These tools and technologies were selected to ensure accuracy, efficiency, and a seamless user experience.

IV. RESULTS

The Waste2Worth system was developed as a high-fidelity prototype with all the main functions of our application, which includes smart waste submission, points computation, finding a nearby station and solving eco-challenges. The prototype demonstrated high usability and supported efficient task completion. This demonstrated that the prototype has captured the main objectives of our Waste2Worth System's core vision (See Fig. 4).

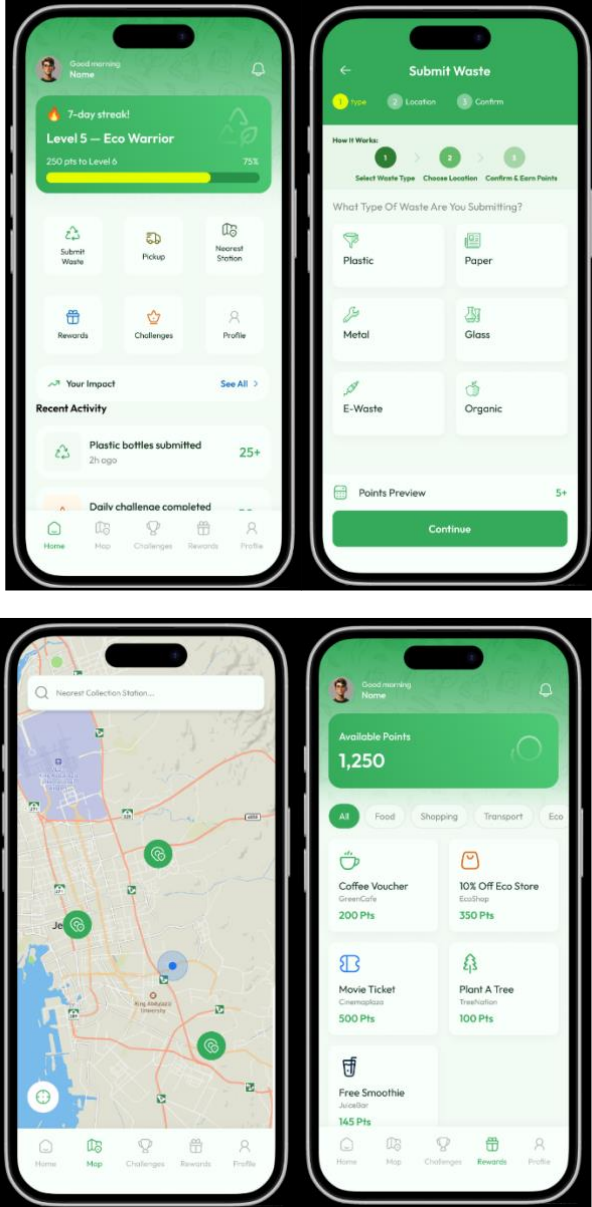


Fig. 4. Final High-Fidelity Prototype Screens of Waste2Worth.

The usability evaluation produced overall positive feedback regarding system functionality and interface usability. This positive feedback came mostly from participants regarding the ease of navigating the app and the ease of finding other features, such as the rewards feature and map feature. None of the participants had difficulties locating stations that were closest to them, disposing of waste, or tracking how many points they had accumulated. Users found that the app flowed in an intuitive manner, which allowed for

an easy change from one feature to another with very little confusion. In addition, users indicated that the gamification elements (badges, streaks, and challenges) in the app increased their desire for participation in the waste management process, thus providing a gratifying experience beyond the intrinsic benefit of the waste management process alone.

According to Table II, all primary tasks of the participants were completed successfully, further supporting evidence of the application's overall usability and accessibility. The UCD approach used in the development of the application has proven effective by enhancing interface organization and pathfinding through user-centered feedback. At the close of the testing phase, Waste2Worth received an 83 on the SUS, meaning good usability and a positive experience which is shown in Fig. 5. Overall, users found the app to be an intuitive way to practice sustainable waste management.

During this stage, the evaluation involved 23 participants divided between usability testing and heuristic evaluation. The usability test with 15 participants focused on usability-related points (i.e. ease of use or other user interface clarity), overall user experience, while the 8 heuristics testcases covered how well our system conforms to established usability standards and/or design guidelines.

Thus, the distribution of these two types of tests indicates a balanced fashion for identifying both the user-facing issues as well as the design-related discrepancies.

Task	Result	Completion Rate
Locate nearby station	Successfully completed	100% (15/15)
Submit recyclable waste	Successfully completed	93% (14/15)
Redeem reward points	Successfully completed	100% (15/15)
Navigate challenges section	Easy for most users	80% (12/15)
Overall SUS Score	82.3 / 100 - Good	-

TABLE II. Summary of Usability Testing Results.



Fig. 5. System Usability Scale Evaluation Result.

V. DISCUSSION

Combining HCI concepts with gamification had a positive influence on recycling behavior and user participation, resulting in a SUS score of 82.3 The application of UCD principles contributed to improved interface usability and navigation efficiency, and gamification elements such as

tracking progress, redeeming rewards, and interactive maps increased the degree to which users engaged with recycling.

Users were motivated to earn points, complete challenges, and keep track of what they had achieved, supporting previous research about gamification and sustainability reward systems. The challenge completion rate (83%) in comparison to simple task completion rate (100%) indicates that there needs to be a better calibration of challenge complexity for future versions to make it easier to access.

IoT integration through intelligent collection systems offers an efficient solution for providing modern waste disposal methods by automating the weighing of waste material and calculating user rewards [3], [5]-[8]. However, the study is still in the high-fidelity prototype phase with no working IoT infrastructure or deployment in live environments currently.

The major achievement of this research is to show how integrating IoT, gamification, rewards, location services, and HCI-based evaluation into one mobile platform is possible and well received by users.

VI. CONCLUSION AND FUTURE WORKS

Waste2Worth presents a user-centered mobile solution that integrates IoT-enabled recycling infrastructure with gamified reward mechanisms, Internet of Things (IoT)-based smart dumpsters and a user rewards system to deliver a positive environmental impact. It leverages Human-Computer Interaction (HCI) and UCD principles to facilitate smart waste submission, reward tracking, eco-challenges and discovery of nearby locations. Usability evaluation yielded a System Usability Scale (SUS) score of 82.3 (good usability) and 100% task completion across all primary tasks indicating that gamifying smart waste management positively influences long-term eco-friendly behavior.

The Waste2Worth project is currently based on a high-fidelity prototype but has results to substantiate further research and development efforts. Future opportunities exist through classification of wastes using Artificial Intelligence (AI) to eliminate manual sorting of materials, the issuance of rewards through blockchain technology to improve transparency/trust and real-world deployments of the application to provide empirical data regarding its value. These opportunities are likely to dramatically increase scalability, durability and overall societal impact on the Waste2Worth platform.

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